

Applied Linear Algebra

Linear Functions

Devi Prasad M
Manipal School of Information Sciences
Manipal

August 2026

We write $f : \mathbb{R}^n \rightarrow \mathbb{R}$ to mean that the function f maps real n -vectors to real numbers.

Function f is a scalar-valued function of n -vectors.

Example

$$f(\mathbf{x} : \mathbb{R}^n) = \mathbf{1}_n^T \mathbf{x}$$

A linear function satisfies the following two properties:

- *Additivity.* $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$ for all n -vectors $\mathbf{x}$ and $\mathbf{y}$.
- *Homogeneity.* $f(\alpha\mathbf{x}) = \alpha f(\mathbf{x})$ for all n -vectors $\mathbf{x}$ and all scalars α .

Prove the following

Definition

A function f is *linear* if it satisfies the *superposition principle*:

$$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y})$$

for all n -vectors $\mathbf{x}$ and $\mathbf{y}$, and all scalars α and β .

Prove the following

Definition

A function f is *linear* if it satisfies the *superposition principle*:

$$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y})$$

for all n -vectors $\mathbf{x}$ and $\mathbf{y}$, and all scalars α and β .

Hint

- Substitute $\alpha \mathbf{x}$ with $\mathbf{u}$ and $\beta \mathbf{y}$ with $\mathbf{v}$.
- Apply *additivity* and *homogeneity* to arrive at the result.

Consider the vector inner product function

$$f(\mathbf{x}) : \mathbb{R} = \mathbf{a}^T \mathbf{x} = a_1 x_1 + \cdots + a_n x_n$$

where $\mathbf{a}$ is a fixed n -vector, and $\mathbf{x}$ is a variable n -vector.

The function f is called a *linear function* of $\mathbf{x}$.

The vector $\mathbf{a}$ is called the *coefficient vector* of the linear function f .

$\mathbf{a}$ is also called the *weight vector* of the linear function f .

Prove the following

The vector inner product function

$$f(\mathbf{x}) : \mathbb{R} = \mathbf{a}^T \mathbf{x} = a_1 x_1 + \cdots + a_n x_n, \quad \mathbf{a} : \mathbb{R}^n, \quad \mathbf{x} : \mathbb{R}^n$$

is linear.

The vector inner product function

$$f(\mathbf{x}) : \mathbb{R} = \mathbf{a}^T \mathbf{x} = a_1 x_1 + \cdots + a_n x_n, \quad \mathbf{a} : \mathbb{R}^n, \quad \mathbf{x} : \mathbb{R}^n$$

is linear.

Proof

Given the inner product function $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$, we can verify that it is linear:

$$\begin{aligned} f(\alpha \mathbf{x} + \beta \mathbf{y}) &= \mathbf{a}^T (\alpha \mathbf{x} + \beta \mathbf{y}) \\ &= \mathbf{a}^T (\alpha \mathbf{x}) + \mathbf{a}^T (\beta \mathbf{y}) \\ &= \alpha (\mathbf{a}^T \mathbf{x}) + \beta (\mathbf{a}^T \mathbf{y}) \\ &= \alpha f(\mathbf{x}) + \beta f(\mathbf{y}) \end{aligned}$$

Definition

The linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ with scalars $c_1, c_2, \dots, c_n$ is the vector

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \sum_{i=1}^n c_i\mathbf{v}_i$$

The scalars are called the *weights* of the linear combination.

By definition, a linear combination involves only finitely many vectors.

Definition (Linear Independence)

A set of vectors where no single vector can be written as a linear combination of the others.

Definition (Span)

The complete set of all possible vectors one can reach by changing the scalar values in a linear combination.

A *linear combination* has the form

$$\sum_{i=1}^k c_i \mathbf{v}_i,$$

with no restriction on the coefficients.

An *affine combination* has the additional constraint

$$\sum_{i=1}^k c_i = 1.$$

The coefficients may be positive, zero, or negative.

In an *affine combination* $\sum_{i=1}^k c_i = 1$, the coefficients may be positive, zero, or negative.

In addition, if

$$c_i \geq 0 \quad \text{for all } i,$$

then the affine combination is called a *convex combination*.

An n -vector $\mathbf{x}$ is a *feature vector* of n features.

A *regression model* is the affine function of $\mathbf{x}$ given by

$$\hat{\mathbf{y}} = \mathbf{x}^T \boldsymbol{\beta} + \mathbf{v}$$

where $\boldsymbol{\beta}$ is an n -vector and $\mathbf{v}$ is an n -vector.

The entries of $\mathbf{x}$ are called the *regressors*.

$\hat{\mathbf{y}}$ is called the *prediction*.

Prediction of Intensive Care Unit Length of Stay in the MIMIC-IV Dataset

<https://www.mdpi.com/2076-3417/13/12/6930>

”Four different classification and regression models for predicting LOS in the ICU are utilized and their performances are compared. The first is the common logistic/linear regression model (for classification/regression problems), which incorporates a linear combination of all features in modeling. ...”

features: body temperature, chloride, creatinine, heart rate, mean corpuscular volume (MCV), O2 saturation, and respiratory rate